

Fig. 3. General outline for Read and Write access to the database.

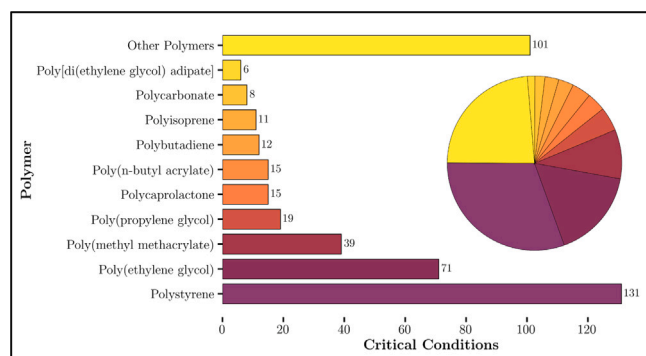


Fig. 4. Ten most common polymers present in the *PolyCrit* database.

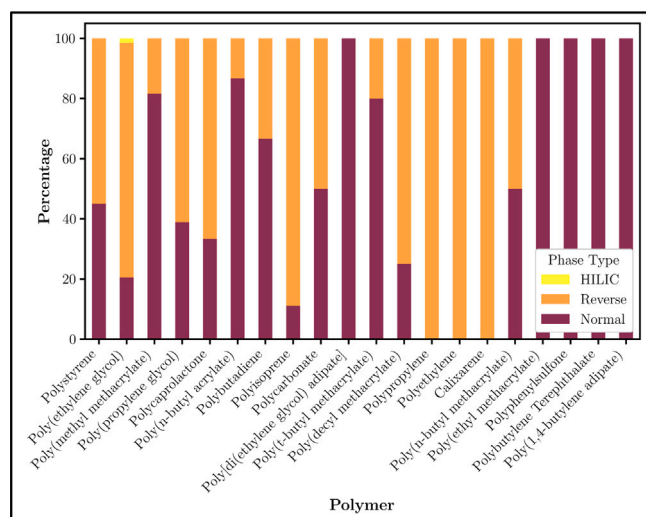


Fig. 5. Relative percentages of stationary phase type for the twenty most common polymers in *PolyCrit*.

are sufficient to prove a set of conditions as critical. Consequently, more data points provided for an experiment should be reflected in the data's reliability; thus, each critical condition was graded with a Reliability score (Table 2).

Nonetheless, various other factors may adversely affect the data, so each entry is assigned an overall Quality score to account for these influences. The Quality score starts at ten and loses a point for every negative quality present in an entry:

- indirect Reliability score (special reduction: -2)
- low Reliability score
- missing "Confirmed Critical Range"
- missing "Particle Diameter" of stationary phase
- missing "Pore Size" of stationary phase
- missing "Column Dimensions" of stationary phase

- unknown stationary phase "phase" (normal, reverse, etc.)
- unable to access paper (by our team at the point of this publication)
- had to be translated to English
- critical conditions present in only one paper

While a lower Quality rating does not imply any inaccuracy in the associated entry, it serves as a prompt for viewers to conduct further investigation. Each relevant entry includes at least one DOI hyperlink for convenient access to the resource, although some older resources or those from certain publication mediums may lack a DOI. To facilitate seamless access to these resources and support deeper investigation, we implemented a structured, user-friendly interface designed for researchers into the website.

3.4. Database access permissions

The website was designed to serve as a dynamic, user-friendly platform that researchers can easily navigate. To begin, all guests to the website can access and browse the database once they've signed in through email and become users, allowing interested parties to explore the curated data (Fig. 3). Additionally, this process allows us to keep track of members of the field who are utilizing our database. The website's structure is designed to be intuitive, enabling visitors to quickly find the information they need and to draw meaningful conclusions from the data. Additionally, the database is equipped with advanced search and filter options, making it possible to narrow down results based on specific criteria or combinations of conditions.

We acknowledge that some papers containing critical conditions were likely missed in this review, and we hope researchers will help us by reporting their work directly to the website. For such researchers wishing to contribute their own data to the database, they must contact a site administrator to register an account. In this process, the credentials of each contributing researcher will be verified, ensuring that all submissions come from reputable sources.

After verification, the researcher gains access to the polymer entry page, a dedicated area of the website where they can submit detailed information about their peer-reviewed LCCC conditions. Here, researchers are guided to enter specific parameters aligned with the standards outlined in Fig. 2, ensuring consistency and clarity across all submissions.

Finally, each new entry undergoes a review by an administrator before being added to the website's database. During this review, the administrator checks both the content of the submission and its accompanying peer-reviewed paper to confirm its validity.

4. Results and discussion

To better present the breadth of literature compiled in *PolyCrit*, we have produced several figures. Fig. 4 illustrates the distribution of unique critical conditions across the ten most common polymers in the *PolyCrit* database (though it currently contains 64 unique polymers). This figure highlights the diversity in critical condition data available for each polymer type, with Polystyrene having the largest number of unique conditions at 131, followed by Poly(ethylene glycol) with 71 conditions. In contrast, polymers like Polybutylene Terephthalate and Poly(ethyl methacrylate) have only a few critical conditions listed, which could indicate either limited application or a lack of comprehensive studies for these polymers. The diversity in the number of conditions across polymers highlights the value of *PolyCrit* as a consolidated resource, where underrepresented polymers can be more readily identified, potentially guiding future research efforts.

Fig. 5 provides a valuable overview of the distribution of stationary phase types — Normal, Reverse and Hydrophilic Interaction Chromatography (HILIC) — across the 20 most common polymers in the *PolyCrit* database. In Normal phases, the column substrate is polar